

1. Let V be a finite-dimensional inner product space over $\mathbb{C}$, and let T be a linear transformation on V . Prove that T is self-adjoint if and only if $(\mathbf{x}, T\mathbf{x})$ is real a real number for all $\mathbf{x} \in V$.

Proof " $\Rightarrow$ " Suppose $T^* = T$. b/c $T^* = T$

$$\overline{(\vec{x}, T\vec{x})} \stackrel{\text{Axiom 2 of inner product}}{=} (T\vec{x}, \vec{x}) \stackrel{\text{by definition of } T^*}{=} (\vec{x}, T^*\vec{x}) \stackrel{\text{b/c } T^* = T}{=} (\vec{x}, T\vec{x})$$

$\Rightarrow (\vec{x}, T\vec{x}) \in \mathbb{R}$ as the conjugate of $(\vec{x}, T\vec{x})$ is itself.

" $\Leftarrow$ " $\forall \vec{x} \in V$: $\overline{(\vec{x}, T\vec{x})} \stackrel{\text{by Axiom 2 of inner product}}{=} (T\vec{x}, \vec{x}) \stackrel{\text{by definition of } T^*}{=} (\vec{x}, T^*\vec{x})$

On the other hand, as $(\vec{x}, T\vec{x}) \in \mathbb{R}$, $\overline{(\vec{x}, T\vec{x})} = (\vec{x}, T\vec{x})$

$$\Rightarrow (\vec{x}, T^*\vec{x}) = (\vec{x}, T\vec{x}) \text{ for all } \vec{x} \in V$$

if $(\vec{x}, \vec{y}) = (\vec{x}, \vec{z})$
for all $\vec{x} \in V$
then $\vec{y} = \vec{z}$.

$$\Rightarrow T^*\vec{x} = T\vec{x} \text{ for all } \vec{x} \in V$$

$$\Rightarrow T = T^*$$

$\Rightarrow T$ is self-adjoint.

2. Let $A \in \mathbb{C}^{n \times n}$. Prove that A is normal if and only if A can be written in the form of $A = A_1 + iA_2$ where A_1 and A_2 are Hermitian and $A_1A_2 = A_2A_1$.

(Hint: Let $A_1 = \frac{1}{2}(A + A^*)$ and $A_2 = \frac{1}{2i}(A - A^*)$.)

Proof " $\Rightarrow$ ". Suppose A is normal. Define $A_1 = \frac{1}{2}(A + A^*)$ $A_2 = \frac{1}{2i}(A - A^*)$.

$$\text{Then } A_1 + iA_2 = \frac{1}{2}(A + A^*) + i \cdot \frac{1}{2i}(A - A^*) = \frac{1}{2}(A + A^*) + \frac{1}{2}(A - A^*) = A$$

$$A_1^* = \left[\frac{1}{2}(A + A^*) \right]^* = \frac{1}{2}(A^* + A) = A_1 \Rightarrow A_1 \text{ is Hermitian}$$

$$A_2^* = \left[\frac{1}{2i}(A - A^*) \right]^* = \overline{\left(\frac{1}{2i} \right)} (A^* - A) = -\frac{1}{2i}(A^* - A) = \frac{1}{2i}(A - A^*) = A_2 \Rightarrow A_2 \text{ is Hermitian}$$

$$\text{(Note } \frac{1}{2i} = \frac{i}{2i \cdot i} = -\frac{i}{2} \Rightarrow \overline{\left(\frac{1}{2i} \right)} = \frac{i}{2} = -\frac{1}{2i} \text{)}$$

$$A_1A_2 = \frac{1}{2}(A + A^*) \cdot \frac{1}{2i}(A - A^*) = \frac{1}{4i}(A^2 - AA^* + A^*A - A^*A^*) \stackrel{\text{b/c } AA^* = A^*A}{=} \frac{1}{4i}(A^2 - A^*A^*A)$$

$$A_2A_1 = \frac{1}{2i}(A - A^*) \cdot \frac{1}{2}(A + A^*) = \frac{1}{4i}(A^2 + AA^* - A^*A - A^*A^*) \stackrel{\text{b/c } AA^* = A^*A}{=} \frac{1}{4i}(A^2 - A^*A^*A)$$

$$\Rightarrow A_1A_2 = A_2A_1$$

" $\Leftarrow$ ": $A^*A = (A_1 + iA_2)^* (A_1 + iA_2) \stackrel{\text{b/c } A_1, A_2 \text{ Hermitian}}{=} (A_1^* - iA_2^*) (A_1 + iA_2) = (A_1 - iA_2)(A_1 + iA_2)$

$$= A_1^2 + iA_1A_2 - iA_2A_1 + A_2^2 \stackrel{A_1A_2 = A_2A_1}{=} A_1^2 + A_2^2$$

$$AA^* = (A_1 + iA_2)(A_1 + iA_2)^* \stackrel{\text{1 of 6}}{=} (A_1 + iA_2)(A_1^* - iA_2^*) = (A_1 + iA_2)(A_1 - iA_2) = A_1^2 + iA_2A_1 - iA_1A_2 + A_2^2$$

$$= A_1^2 + A_2^2$$

$$\Rightarrow A^*A = AA^*$$

3. Prove that a normal and nilpotent linear transformation is the zero linear transformation.
(Note: A linear transformation T is nilpotent if there exists a positive integer r such that $T^r = 0$.)

Proof: Suppose T is a normal and nilpotent linear transformation.

By Spectral Theorem, there exists an orthonormal basis such that $[T]_{\mathcal{B}} = D$ is a diagonal matrix. Denote $D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$

As T is nilpotent, $\exists r \in \mathbb{Z}^+$ such that $T^r = 0$

$$\Rightarrow 0 = [T^r]_{\mathcal{B}} = ([T]_{\mathcal{B}})^r = D^r = \begin{pmatrix} \lambda_1^r & & \\ & \ddots & \\ & & \lambda_n^r \end{pmatrix}$$

$$\Rightarrow \lambda_1^r = \lambda_2^r = \dots = \lambda_n^r = 0 \Rightarrow \lambda_1 = \lambda_2 = \dots = \lambda_n = 0$$

$\Rightarrow [T]_{\mathcal{B}}$ is the zero matrix

$$\Rightarrow T = 0.$$

4. Let $A \in \mathbb{C}^{n \times n}$ be a Hermitian matrix. Prove that A is positive definite if and only if all the eigenvalues of A are positive.

By Spectral Theorem, since A is Hermitian, there exists a unitary matrix $P = (\vec{p}_1 \vec{p}_2 \dots \vec{p}_n) \in \mathbb{C}^{n \times n}$.

and a diagonal matrix $D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \in \mathbb{R}^{n \times n}$ such that $A = PDP^*$

$$\Rightarrow AP = PD \Rightarrow A(\vec{p}_1 \vec{p}_2 \dots \vec{p}_n) = (\vec{p}_1 \vec{p}_2 \dots \vec{p}_n) \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$$

$$\Rightarrow (A\vec{p}_1 \ A\vec{p}_2 \ \dots \ A\vec{p}_n) = (\lambda_1\vec{p}_1 \ \lambda_2\vec{p}_2 \ \dots \ \lambda_n\vec{p}_n)$$

$$\text{i.e. } A\vec{p}_i = \lambda_i \vec{p}_i \text{ for } i=1, 2, \dots, n.$$

$\Rightarrow \lambda_1, \lambda_2, \dots, \lambda_n$ are all the eigenvalues of A .

As P is unitary $\{\vec{p}_1, \vec{p}_2, \dots, \vec{p}_n\}$ is an orthonormal basis of $\mathbb{C}^n$. i.e. $\vec{p}_i^* \vec{p}_j = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$

" $\Rightarrow$ " If A is positive definite, then $\forall \vec{x} \neq 0, \vec{x}^* A \vec{x} > 0$

In particular, $0 < \vec{p}_i^* A \vec{p}_i = \vec{p}_i^* \lambda_i \vec{p}_i = \lambda_i \vec{p}_i^* \vec{p}_i = \lambda_i$ for each $i=1, 2, \dots, n$.

" $\Leftarrow$ " Suppose $\lambda_i > 0$ for all $i=1, 2, \dots, n$,

$\forall \vec{x} \in \mathbb{C}^n$. as $\{\vec{p}_1, \dots, \vec{p}_n\}$ is an orthonormal basis of $\mathbb{C}^n$: $\vec{x} = \sum_{i=1}^n a_i \vec{p}_i$

$$\begin{aligned}
 \text{Then } \vec{x}^* A \vec{x} &= (A \vec{x}, \vec{x}) = \left(A \left(\sum_{i=1}^n a_i \vec{p}_i \right), \sum_{k=1}^n a_k \vec{p}_k \right) \\
 &= \left(\sum_{i=1}^n a_i (A \vec{p}_i), \sum_{k=1}^n a_k \vec{p}_k \right) \\
 &= \left(\sum_{i=1}^n a_i \lambda_i \vec{p}_i, \sum_{k=1}^n a_k \vec{p}_k \right) \\
 &= \sum_{j=1}^n a_j \lambda_j \bar{a}_j (\vec{p}_j, \vec{p}_j) \\
 &= \sum_{j=1}^n \lambda_j |a_j|^2 > 0
 \end{aligned}$$

$\Rightarrow A$ is positive definite.

5. Let $A \in \mathbb{C}^{n \times n}$ be a normal matrix. Prove that A is unitary if and only if every eigenvalue of A has magnitude (or absolute value) 1.

Since A is normal, by Spectral Theorem, there exists a unitary matrix $P \in \mathbb{C}^{n \times n}$ and a diagonal matrix $D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \in \mathbb{C}^{n \times n}$ such that $A = PDP^*$. $\Rightarrow \lambda_1, \dots, \lambda_n$ are all the eigenvalues of A .

$$A \text{ is unitary} \Leftrightarrow A^*A = I \Leftrightarrow (PDP^*)^* PDP^* = I$$

$$\Leftrightarrow P D^* P^* P D P^* = I$$

$$\Leftrightarrow P D^* D P^* = I$$

$$\Leftrightarrow P^* (P D^* D P^*) P = P^* I P = I$$

$$\Leftrightarrow D^* D = I$$

$$\Leftrightarrow \begin{pmatrix} \bar{\lambda}_1 & & \\ & \ddots & \\ & & \bar{\lambda}_n \end{pmatrix} \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} = I$$

$$\Leftrightarrow \begin{pmatrix} |\lambda_1|^2 & & \\ & \ddots & \\ & & |\lambda_n|^2 \end{pmatrix} = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix} \Leftrightarrow |\lambda_1| = \dots = |\lambda_n| = 1$$

6. Let $A \in \mathbb{C}^{n \times n}$ be a Hermitian matrix. Suppose A is both positive definite and unitary. Prove that $A = I$. Hint: You may use conclusions from the above two problems.

Proof. Since A is Hermitian, by Spectral Theorem, there exists a unitary matrix P , and a diagonal matrix $D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \in \mathbb{R}^{n \times n}$ such that $A = PDP^*$.

Note that $\lambda_1, \dots, \lambda_n$ are all the eigenvalues of A .

Since A is positive definite: $\lambda_1 > 0, \dots, \lambda_n > 0$

Since A is unitary: $|\lambda_1| = \dots = |\lambda_n| = 1$.

$$\Rightarrow \lambda_1 = \lambda_2 = \dots = \lambda_n = 1$$

$$\Rightarrow D = I$$

$$\Rightarrow A = PDP^* = PIP^* = I.$$



7. Consider $A = \begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}$.

1). Find the singular value decomposition of A . (Show your computation work, do not use technology.)

2). Find the generalized inverse A^+ .

$$A^T A = \begin{pmatrix} -1 & 1 \\ 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

$$\lambda^2 - 3\lambda + 2 = 0$$

$$\lambda^2 - 3\lambda + 1 = 0$$

$$\det(A - \lambda I) = \det \begin{pmatrix} 2-\lambda & -1 & -1 \\ -1 & 1-\lambda & 0 \\ -1 & 0 & 1-\lambda \end{pmatrix} = (-1) \cdot \det \begin{pmatrix} 1 & -1 \\ 1-\lambda & 0 \end{pmatrix} + (1-\lambda) \det \begin{pmatrix} 2-\lambda & -1 \\ -1 & 1-\lambda \end{pmatrix}$$

$$= (-1)(1-\lambda) + (1-\lambda)(2-3\lambda+\lambda^2)$$

$$= (1-\lambda)(\lambda^2-3\lambda)$$

$$= (1-\lambda) \cdot \lambda(\lambda-3)$$

$\Rightarrow$ eigenvalues of $A^T A$: $\lambda_1 = 3$ (multiplicity = 1) $\Rightarrow \sigma_1 = \sqrt{3}$

$\lambda_2 = 1$ (multiplicity = 1) $\Rightarrow \sigma_2 = 1$

$\lambda_3 = 0$ (multiplicity = 1)

$$A^T A - 3I = \begin{pmatrix} -1 & -1 & -1 \\ -1 & -2 & 0 \\ -1 & 0 & -2 \end{pmatrix} \xrightarrow[\text{reduction}]{\text{row}} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow (A^T A - 3I)\vec{x} = \vec{0}$$

$$\Leftrightarrow \begin{cases} x_1 + 2x_3 = 0 \\ x_2 - x_3 = 0 \end{cases} \Leftrightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_3 \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$$

$$\Rightarrow \vec{y}_1 = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \Rightarrow \vec{v}_1 = \frac{\vec{y}_1}{\|\vec{y}_1\|} = \begin{pmatrix} -2/\sqrt{6} \\ 1/\sqrt{6} \\ 1/\sqrt{6} \end{pmatrix}$$

$$A^T A - I = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} \xrightarrow{\text{row reduction}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow (A^T A - I)\vec{x} = \vec{0}$$

$$\Leftrightarrow \begin{cases} x_1 = 0 \\ x_2 + x_3 = 0 \end{cases} \Leftrightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_3 \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

$$\Rightarrow \vec{y}_2 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \Rightarrow \vec{v}_2 = \frac{\vec{y}_2}{\|\vec{y}_2\|} = \begin{pmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

$$A^T A - 0I = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \xrightarrow[4 \text{ of } 6]{\text{row reduction}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow (A^T A - 0I)\vec{x} = \vec{0}$$

$$\Leftrightarrow \begin{cases} x_1 - x_3 = 0 \\ x_2 - x_3 = 0 \end{cases} \Leftrightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_3 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\Rightarrow \vec{y}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \Rightarrow \vec{v}_3 = \frac{\vec{y}_3}{\|\vec{y}_3\|} = \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}$$

$$\Rightarrow V = \begin{pmatrix} -\frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{pmatrix} \quad \Sigma = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\text{Rank } A = 2$$

$$\Rightarrow V_n = \begin{pmatrix} -\frac{1}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{pmatrix} \quad D = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \end{pmatrix}$$

$$\vec{u}_1 = \frac{1}{\sigma_1} A \vec{v}_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} \frac{3}{\sqrt{6}} \\ -\frac{3}{\sqrt{6}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\vec{u}_2 = \frac{1}{\sigma_2} A \vec{v}_2 = \frac{1}{1} \begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$U = U_r = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\Rightarrow \text{SVD of } A: A = U \Sigma V^* \quad \text{with } U, \Sigma, V \text{ defined above}$$

$$\text{Reduced SVD of } A: A = U_r D V_r^* \quad \text{with } U_r, D, V_r \text{ defined above}$$

$$2). \quad A^T = V_r D^T U_r^* = \begin{pmatrix} -\frac{1}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{3}} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{1}{\sqrt{18}} & 0 \\ \frac{1}{\sqrt{18}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{18}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{1}{3} & \frac{1}{3} \\ -\frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{pmatrix}$$

8. Let $A \in \mathbb{C}^{n \times n}$ be a Hermitian and indefinite (i.e. not positive definite or positive semi-definite). Suppose $A = PDP^*$ for some unitary matrix P and some diagonal matrix $D \in \mathbb{R}^{n \times n}$. Find the singular value decomposition of $A = U\Sigma V^*$ by constructing V , Σ and U using some possible variations of the columns or entries of P and D .

Suppose $A = PDP^*$ with $P \in \mathbb{C}^{n \times n}$ unitary, $D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \in \mathbb{R}^{n \times n}$ diagonal.

Without loss of generality, we may assume $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_r| > 0, \lambda_{r+1} = \dots = \lambda_n = 0$.

Denote $P = (\vec{p}_1 \vec{p}_2 \dots \vec{p}_n)$. $A^*A = (PDP^*)^* PDP^* = PD^*P^*PDP^* = PD^*DP^* = PD^2P^*$.

$D^2 = \begin{pmatrix} \lambda_1^2 & & \\ & \lambda_2^2 & \\ & & \ddots \\ & & & \lambda_n^2 \end{pmatrix} \Rightarrow \{\vec{p}_1, \dots, \vec{p}_n\}$ is an orthonormal eigenbasis for A^*A with eigenvalues $\lambda_1^2, \dots, \lambda_n^2$.

$\Rightarrow$ Define $V = (\vec{p}_1 \vec{p}_2 \dots \vec{p}_n)$, $\sigma_1 = |\lambda_1|$, $\sigma_2 = |\lambda_2|$, $\dots$, $\sigma_r = |\lambda_r|$

Then $\Sigma = \begin{pmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r & & \\ & & & 0 & \dots & 0 \end{pmatrix}$.

Define: $\vec{u}_i = \frac{1}{\sigma_i} A \vec{v}_i = \frac{1}{\sigma_i} \lambda_i \vec{v}_i = \begin{cases} \vec{v}_i, & \text{if } \lambda_i > 0 \\ -\vec{v}_i, & \text{if } \lambda_i < 0. \end{cases}$ for $i = 1, \dots, r$

$\vec{u}_j = \vec{v}_j$ for all $j = r+1, \dots, n$.

Define $U = (\vec{u}_1 \vec{u}_2 \dots \vec{u}_n)$ then $U \in \mathbb{C}^{n \times n}$ is unitary

$\Rightarrow A = U\Sigma V^*$

9. Let A be an $m \times n$ matrix, and let P and Q be $m \times m$ and $n \times n$ unitary matrices. Show that A and PAQ have the same singular values.

Proof: $(PAQ)^*(PAQ) = Q^*A^*P^*PAQ = Q^*A^*AQ$

$$\Rightarrow A^*A \text{ is similar to } (PAQ)^*(PAQ)$$

$$\Rightarrow A^*A \text{ and } (PAQ)^*(PAQ) \text{ have exactly the same eigenvalues.}$$

Since the singular values of a matrix B are square roots of eigenvalues of B^*B .

$$\Rightarrow A \text{ and } PAQ \text{ have the same singular values.}$$